COMPUTER NETWORKS — WEEK 1 CHEAT SHEET

Tanenbaum & Wetherall, 6th Ed. | Instructor: Dr. Mohammad Shahidul Islam

1. What is a Computer Network?

A network connects **2 or more autonomous computers** so they can talk and share things. Autonomous means each computer can work alone — it does not need the others.

Example: Phone, laptop, printer on home Wi-Fi = a small network.

Trick: Network = Devices + Cables/Wireless + Sharing

2. LAN, MAN, WAN

Type	Area	Example
LAN	Room / Building / Campus	University computer lab
MAN	City	Cable TV network in Dhaka
WAN	Country / World	The Internet

LAN is fast (100 Mbps–10 Gbps). WAN is slower — data travels far.

Trick: L→M→W = Small→Medium→Large

3. Applications of Networks

- **Resource sharing:** one printer for many users
- **Information sharing:** files, Google search, hospital records
- **Communication:** email, broadcast, remote computer use
- **Distributed processing:** big task split among many computers (Grid computing)

Daily apps: Email, Web, E-commerce, VoIP, Video call, Messenger, Streaming.

Trick: SHARE + TALK + FIND

4. Network Topology (BSRM)

Topology = shape/layout of the network.

- **Bus:** one main cable, all devices attached. Cheap. Cable breaks = whole network down.
- **Star:** all devices connect to a hub/switch. Most common today. Hub fails = all fail.
- **Ring:** devices in a circle, data passes node to node. Single ring = one way; dual ring = two ways (safer).
- **Mesh:** every device connects to every other device. Very reliable, very costly. Used in WANs, banks.

Trick: "Be Smart, Ring Mesh" — Bus, Star, Ring, Mesh

5. Network Components (PINCONA)

- **Physical media:** wired (twisted pair, coaxial, fiber) or wireless (Wi-Fi, Bluetooth, 4G/5G)
- **Interconnecting devices:** Hub (sends to all), Switch (sends to right one), Router (joins networks), Access Point, Modem
- **Computers:** Server (gives service, runs 24/7) and Client (asks for service)
- **Networking software:** protocols that control sending/receiving
- **Applications:** browser, email app, Zoom, FTP client

Trick: Server=Waiter, Client=Customer, Router=Traffic police

6. Protocols, Layers, Interfaces

Protocol = rules both sides follow to talk.
Layer = one job in the network; higher layers use lower layer services.
Peers = same layer on two different machines.
Interface = boundary between two layers next to each other.
Protocol stack = one protocol per layer, all layers together.

Connection Types

- **Connection-oriented:** connect → talk → close (like a phone call). Example: TCP
- **Connectionless:** just send, no setup (like posting a letter). Example: UDP

Service Primitives

REQUEST → INDICATION → RESPONSE → CONFIRM

Trick (Primitives): Call (Request) → Ring (Indication) → Answer (Response) → Connected (Confirm)

7. OSI Model — 7 Layers

7. Application — user interface (HTTP, SMTP, FTP)
6. Presentation — format, encryption, compression
5. Session — starts/ends conversations, sync
4. Transport — reliable delivery, splits data (TCP/UDP)
3. Network — routing, best path (IP)
2. Data Link — makes frames, error check (Ethernet)
1. Physical — sends raw bits (cable, Wi-Fi)

Trick (top→down): All People Seem To Need Data Processing
Trick (bottom→up): Please Do Not Throw Sausage Pizza Away

8. TCP/IP Model — 4 Layers

TCP/IP Layer	Same as OSI
Application	App + Presentation + Session (7,6,5)
Transport	Transport (4)
Internet	Network (3)
Network Access	Data Link + Physical (2,1)

OSI = 7 layers, made for learning. TCP/IP = 4 layers, actually used on the real Internet.

Trick: "A Tiny Internet Network" = App, Transport, Internet, Network Access

9. HDLC (Data Link Control)

HDLC packs data into **frames** and sends them safely between two points.

Frame Part	Job
Flag (01111110)	Marks start/end of frame
Address	Which terminal it is for
Control	Sequence number, ACK info
Data	Actual message
CRC	Checks for errors

3 Frame Types

- **I-Frame** — sends data; 3-bit SEQ number (up to 7 unacked frames)
- **S-Frame** — control only: RR (ready), REJ (resend all), RNR (busy), SREJ (resend one)
- **U-Frame** — connect/disconnect/reset (DISC, SABM, FRMR)

Trick: "Friendly ACDs Cheerfully Farewell" = Flag, Address, Control, Data, CRC, Flag

Quick Exam Recall

- Network = 2+ autonomous computers sharing resources
- LAN < MAN < WAN in size; Internet = biggest WAN
- 4 topologies: Bus, Star, Ring, Mesh
- PINCONA = Physical media, Interconnecting devices, Networks(software), Computers, Operating system, Networking Applications
- Protocol = rules; Peer = same layer, diff machine; Interface = layer boundary
- OSI = 7 layers (theory); TCP/IP = 4 layers (real use)
- HDLC frame = Flag-Address-Control-Data-CRC-Flag

বাংলা সংকলন: নেটওয়ার্ক মানে একাধিক কম্পিউটার একসাথে সংযুক্ত হয়ে তথ্য ও রিসোর্স শেয়ার করা। LAN ছোট এলাকার, MAN শহরের, WAN দেশ/বিশ্বের নেটওয়ার্ক। টপোলজি ৪ রকম: Bus, Star, Ring, Mesh। OSI মডেলে ৭টি স্তর, আর TCP/IP মডেলে ৪টি স্তর থাকে। HDLC ডেটাকে ফ্রেমে ভাগ করে নিরাপদে পাঠায়।